

TELCO CUSTOMER CHURN PREDICTION AND ANALYSIS



Introduction

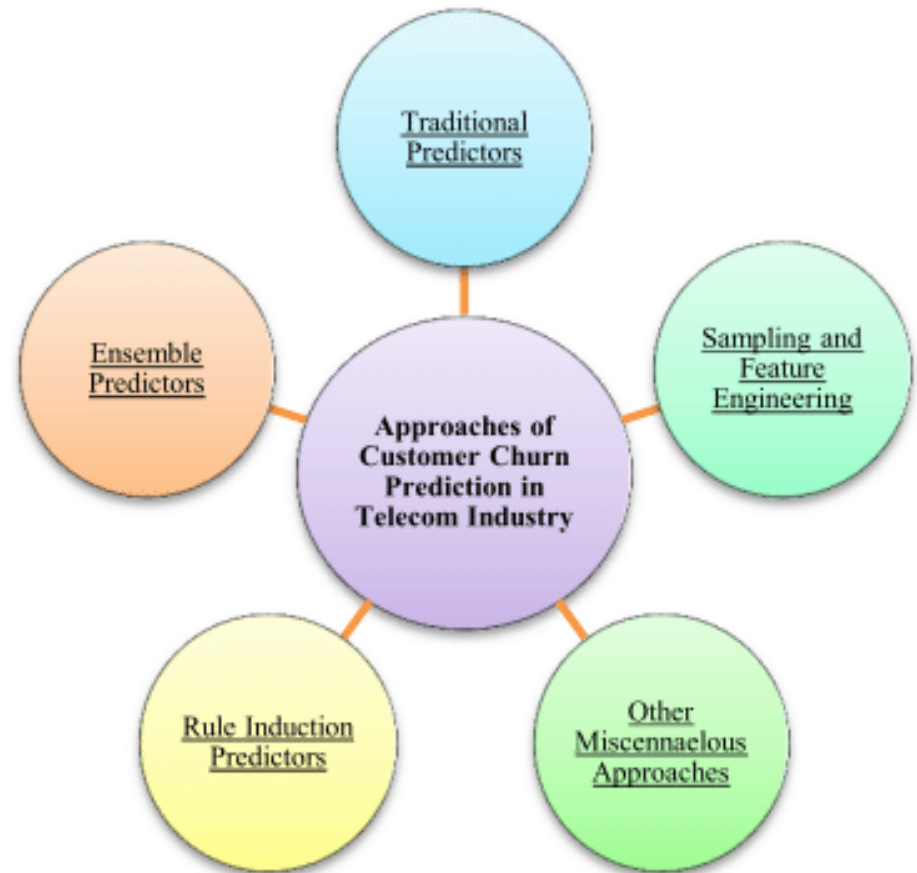
Customer churn is defined as when customers or subscribers discontinue doing business with a firm or service.

Customers in the telecom industry can choose from a variety of service providers and actively switch from one to the next. The telecommunications business has an annual churn rate of 15-25 percent in this highly competitive

Customer churn is a critical metric because it is much less expensive to retain existing customers than it is to acquire new customers.

Problem Statement

- High churn reduces revenue.
- Need early identification.
- Build a prediction model.



Objectives

Analyze customer data

- Train ML model
- Predict churn
- Improve retention

Dataset

Telecom customer dataset
Features: Gender, Tenure,
Charges, Contract,
Payment Method
Target: Churn

Customer Data Analysis

NEW SESSIONS

58 +62%

UNASSIGNED SESSIONS

21 +82%

FIRST RESPONSE

3 _{min} +5%

Regular Customers And New Customers

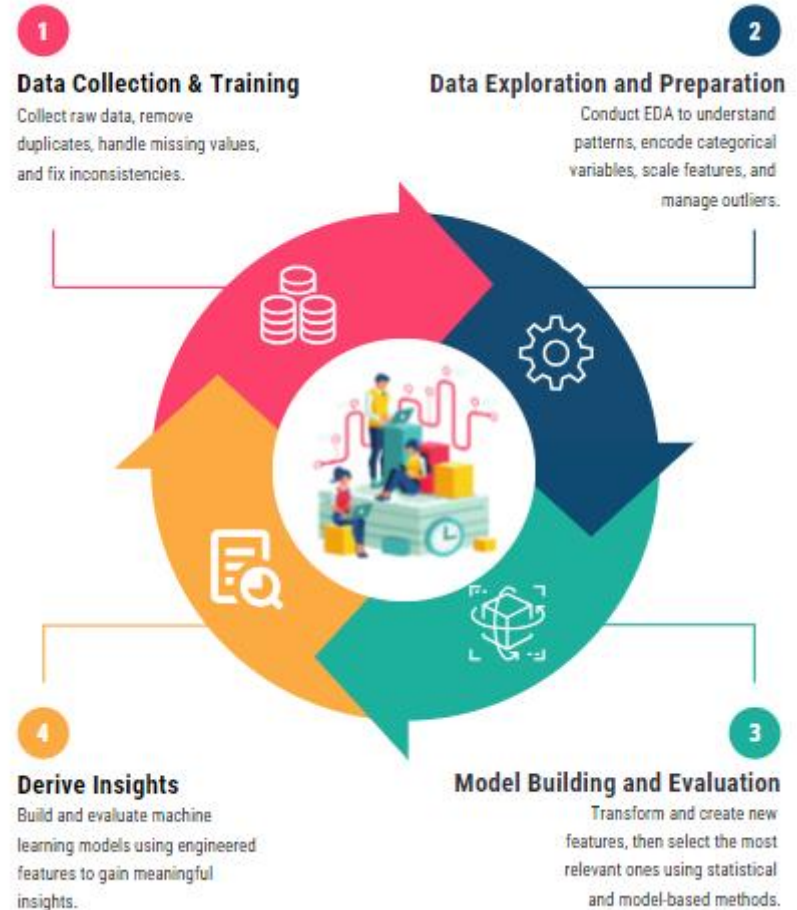
New - 27%

Returning - 73%



Workflow

Data Collection → Cleaning → EDA
→ Feature Engineering → Model
Training → Prediction → Evaluation

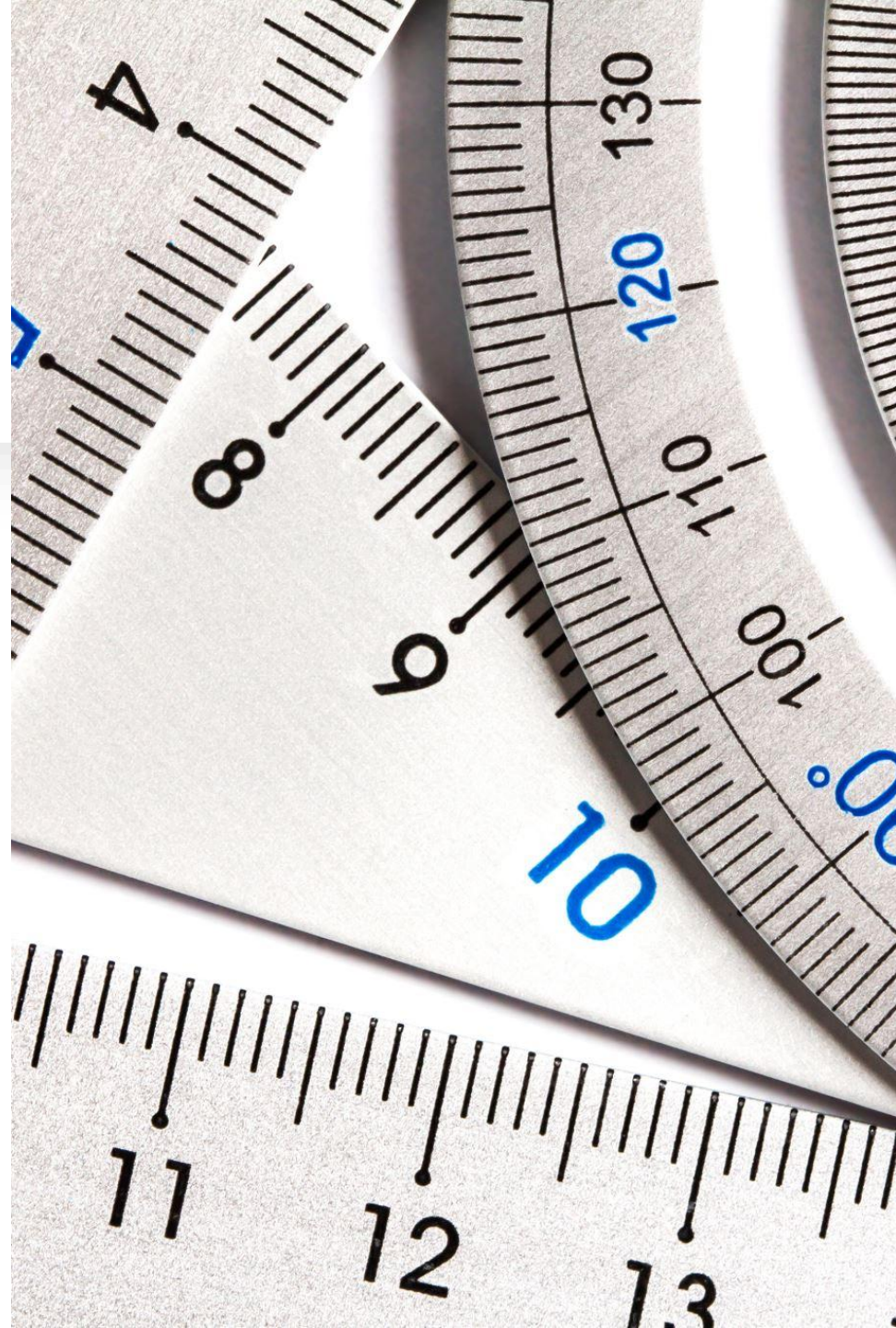


Algorithms

- Logistic Regression
- Decision Tree
- Random Forest
- SVM

Evaluation

- Metrics: Accuracy, Precision, Recall, F1-score, Confusion Matrix
Example Accuracy: 85%



Advantages

- Reduce revenue loss
- Better customer retention
- Improved business decisions

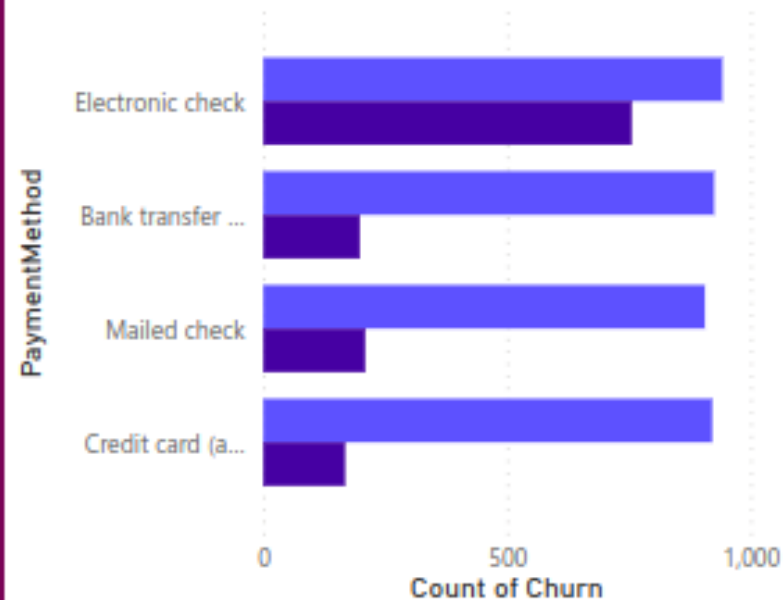


Future Scope

- Deep Learning
- Real-time prediction
- CRM integration

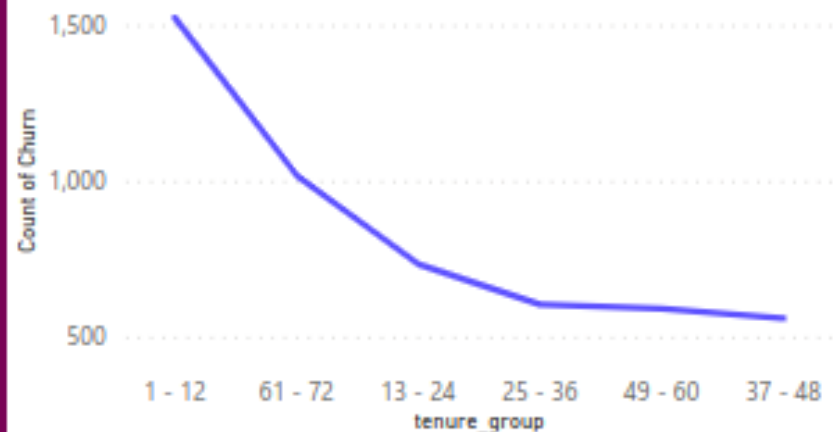
Count of Churn by PaymentMethod and Churn

Churn ● No ● Yes



Average of MonthlyCharges	Average of TotalCharges	Churn
61.53	2,571.17	No
75.29	1,572.20	Yes
65.17	2,306.57	

Count of Churn by tenure_group



Conclusion

- Customer churn prediction enables proactive customer retention and business growth.

